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Femtosecond laser bonding of transparent polycarbonate: a study on the weld seam quality and strength

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Outline

Abstract

Reliable bonding of transparent polymers is essential for microfluidic and biomedical devices. Ultrashort laser welding provides localized energy deposition with minimal heat-affected zones, but nonlinear absorption introduces challenges in achieving consistent weld quality. This study investigates femtosecond laser bonding of transparent polycarbonate using a 1030 nm source, focusing on the effects of fluence per pulse, scanning speed, and repetition rate. Two degradation mechanisms are identified: excessive cumulative fluence and high per-pulse fluence, both defining operational limits. At optimized fluence values, pre-focal absorption before the focus causes a focal offset between the intended and actual weld location. Even after refocusing on the interface, the gap between plates and the distorted beam profile produce a non-uniform fluence distribution, compromising weld quality. Simulations confirm that such offsets distort beam shape and fluence delivery. To overcome this, a low-fluence with high-overscan strategy is proposed, mitigating offset effects. Uniform welds are achieved at a repetition rate of 1 MHz with scanning speeds ranging from 10 to 30 mm/s and fluences of 0.08–0.17 J/cm^2, with more than five overscans. FTIR measurements are performed to monitor potential chemical changes in the bonded regions. The optimized process has a maximum shear strength of 30.2 ± 5 N/mm^2, corresponding to 48% of pristine polycarbonate. Weibull analysis identifies a highest modulus of 5.2, confirming process reliability. Devices withstand leakage tests up to 1 bar. This approach establishes a reproducible femtosecond welding strategy for transparent polymers, resulting in stronger, and more reliable fabrication of microfluidic and biomedical systems.

Keywords

Femtosecond laser; Laser welding; Non-linear absorption; Polycarbonate

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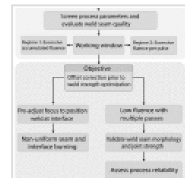
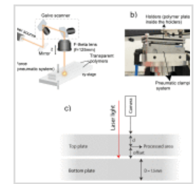
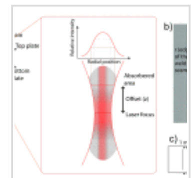
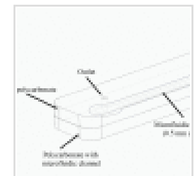
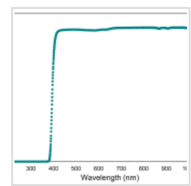
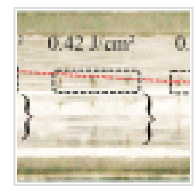
Nomenclature

Symbol/term	Description	Unit
λ	Laser wavelength	nm
E	Pulse energy	$\mu\mathrm{J}$
F_0	Peak fluence at the focus	$\mathrm{J/cm^2}$
F_{Cum}	Cumulative fluence	$\mathrm{J/cm^2}$

Figures (21)

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Tables (4)

ω_0	Beam waist (radius at $1/e^2$)	μm
$\omega(z)$	beam radius at distance z	μm
Z_R	Rayleigh's range	μm
M^2	Beam quality factor	–
NA	Numerical aperture of the focusing lens	–
n_{eff}	Effective number of pulses	–
OL	Pulse overlap	%
z	Vertical offset	mm
L	Weld line length	mm
r	Weld seam radius	μm
A	Bonded area	mm^2
F	Breaking force	N
τ	Shear strength	N / mm^2
R^2	Coefficient of determination (Weibull fit)	%
β	Weibull modulus	–
PC	Polycarbonate	–
$UV-VIS-NIR$	Ultraviolet–Visible–Near Infrared spectroscopy	–
$FTIR$	Fourier-Transform Infrared spectroscopy	–
$FWHM$	Full Width at Half Maximum	–

1. Introduction

Polymer bonding plays a crucial role in microfluidic chip fabrication, especially for transparent polymers used in biomedical applications [1], [2]. Microfluidic chips are often composed of two or more transparent polymer plates that contain microscale channel structures. Ensuring a flawless bond between plates is crucial to create enclosed pathways that enable accurate fluid movement [3]. Therefore, high-precision bonding methods are required during the manufacturing of microfluidic chips to effectively prevent both intra- and inter-channel leaks [4].

There are two primary categories of bonding: indirect and direct bonding. Indirect bonding involves the use of supplementary materials, such as chemical reagents or tapes, to create bonds. Within thermoplastic bonding, indirect methods include adhesive bonding, chemical bonding (surface modification), and microwave bonding. Adhesive bonding involves the use of an adhesive layer between two thermoplastic plates [5]. Adhesives can exist in either dry or liquid form. This method's primary advantages are its simplicity, optical clarity, ability to join dissimilar materials, gas permeability for cell culture applications, and compatibility with pre-bonding (bio)functionalisation [6], [7]. However, concerns include cytotoxicity, adsorption, and the need for post-processing like UV curing or plasma treatment [8], [9], [10]. Dry adhesives may limit channel thickness, and liquid adhesives risk the deformation or clogging of channels due to reflow [11]. Another indirect bonding method is microwave bonding, in which microwaves are used to heat a conductive intermediate layer placed between two thermoplastic chip halves. This localized heating reduces deformation and clogging risks and requires relatively inexpensive equipment. However, depositing the conductive layer adds complexity to chip design [12]. In chemical bonding, the polymer surface is activated at the interface with functional groups, allowing them to bond through strong covalent interactions [13]. This method enables bonding at low temperatures and pressures, preserving the geometry of the microchannels [14]. Nevertheless, the multiple processing steps involved limit its commercial scalability. Femtosecond bonding with nanoparticles is another form of indirect bonding, where a metallic or composite interlayer is introduced to enhance energy absorption. For example, silver nanofilms have been used to achieve high-quality femtosecond-laser welding of glass by promoting localized plasma formation and improved interfacial heating [15]. Ultrafast laser studies further show that such interlayers can support strong bonding in transparent materials when higher absorption is needed [16]. Similar concepts appear in low-temperature sintering of graphene–Cu nanoparticles, which provide strong joints in electronic and composite systems [17]. However, these approaches require a deposited metallic layer, reduce optical transparency, and introduce potential contamination, making them unsuitable for transparent, biocompatible polymer microfluidic bonding [15].

The direct bonding category includes techniques that do not require additional materials or reagent layers at the interface. Direct bonding methods include thermal bonding, solvent bonding, ultrasonic welding, and laser welding. Thermal bonding involves heating polymers above their glass transition temperature (T_g) and pressing them together to allow inter-diffusion [18]. This method is fast and produces strong bonds but requires high temperatures and pressures, which can deform microchannels and prevent pre-bonding functionalisation [19]. Solvent bonding is a direct method in which a solvent, either in liquid or vapor form, is applied to the polymer surfaces to dissolve the top layers and mobilize polymer chains. When pressed together, the chains merge as the solvent evaporates [20]. This approach is rapid, cost-effective, and maintains optical clarity. However, excess solvent may cause channel clogging or distortion, and uneven evaporation can cause weak or inconsistent bonding [21]. Another drawback of this method is the need for additional steps, like plasma treatment or UV irradiation [22]. Ultrasonic welding involves locally melting the polymer by propagating ultrasonic vibrations between a weld horn and an anvil [23]. The generated mechanical vibrations are concentrated at the interface of the polymer substrates, producing enough heat through friction to fuse the thermoplastics [24]. It is a fast process with a short cycle time, suitable for mass production [25]. However, polymer shrinkage may limit channel height, and uneven energy distribution can cause insufficient or excessive fusion, resulting in an inhomogeneous weld seam [12]. Additionally, melted polymer can reflow into the channels, causing deformation and clogging [12]. A common challenge across all these methods is the limited precision, which is especially critical for microfluidic applications. Laser-based welding, however, offers a precise, heat-controlled solution for bonding chip halves without compromising material integrity.

Laser welding is a precise, direct bonding method that can be used for thermoplastic substrates as well as other materials such as glasses, metals, and composites. In this process, polymer plates are held close together by applying force from the top and bottom. The laser light passes through one plate to reach the interface, where it is absorbed. A key limitation is the requirement for the top polymer plate to be transparent to the laser wavelength. The absorbed light generates enough heat at the interface to bond the two thermoplastic substrates. The light absorption mechanism divides laser welding into two categories: transmission laser welding and direct laser welding [26], [27]. Transmission laser welding works by allowing the laser beam to pass through the top transparent polymer plate, where it is then absorbed by the bottom layer [28]. To facilitate this, the lower plate must either be intrinsically absorptive to the incoming laser light or doped with an absorber (e.g., carbon black, dyes, or additives) [29], [30]. This requirement imposes limitations on material selection, often forcing the use of opaque bottom substrates. This not only limits compatibility between different polymer types but also makes visual monitoring of the liquid flow more difficult. An alternative approach is transmission laser welding using a thulium-doped fiber laser emitting at 2 μm, where the low absorption coefficient of many polymers enables light to propagate through the full thickness. Energy deposition then occurs throughout the material, allowing absorber-free welding [31]. In contrast, direct laser welding, enabled by ultrashort pulsed lasers, allows bonding between two transparent polymers via nonlinear absorption at the interface [32]. The high localized fluence excites electrons in the valence band to the conduction band through multiphoton absorption. Once in the conduction band, these electrons can gain additional energy from the laser field and collide with bond electrons, ionizing them through initial impact and subsequently via avalanche ionization, a chain reaction that rapidly increases the free electron density in the focal volume. As a result, a dense plasma is formed, which absorbs additional laser energy through inverse Bremsstrahlung, leading to rapid energy deposition and localized heating via electron-phonon coupling. The laser-induced heating is expected to raise the local temperature in the processed region above the glass transition temperature (T_g) of polycarbonate, causing the amorphous phase to soften and increasing segmental chain mobility, which enables interpenetration across the interface. Upon cooling below T_g, the polymer stiffens again, and the entangled chains become locked in place, resulting in a stable weld [33]. Since this process is confined to the laser's focal point, the surrounding material remains unaffected, preserving transparency. Additionally, no absorbing layers are required, and the minimized heat-affected zone is another significant advantage of using ultrashort laser pulses [34]. Advancements in direct laser welding have demonstrated great promise in bonding transparent polymers, particularly for microfluidic applications. For instance, Volpe et al. successfully bond transparent PMMA plates using 1030 nm, 650 fs pulses operating at 5 MHz, producing welds that held up to 1 bar of internal pressure [32]. Mizuguchi et al. achieved strong bonds by shaping the energy distribution dendritically using a 1064 nm laser with 250 fs pulses, successfully bridging 14 μm air gaps and achieving shear strengths of 21.4 MPa at 1 mm/s [27]. Roth et al. [37] extended this research by exploring the welding of cyclo-olefin copolymer (COC) using a 1028 nm femtosecond laser operating at 571 kHz. They demonstrated that high-speed welding, up to 20 mm/s in a single pass, is feasible while maintaining weld seam widths between 38 and 137 μm, depending on the laser power. The resulting welds achieved shear strengths up to 40 MPa, confirming the mechanical robustness of the joints [35]. This finding is particularly important for improving process efficiency in industrial settings. In a follow-up study, Roth et al. [35] investigated the influence of welding speed on focal plane stability during ultrashort pulse laser welding of COC. They observed that increasing scanning speeds induces a reverse shift in the focal plane, which can significantly affect weld quality. Specifically, a focal offset, distance between the intended focal position and the actual processing zone, in the range of −0.68 μm to −0.56 μm was found to produce stable and high-quality weld seams at a scanning speed of 25 mm/s. However, at even higher speeds, the available laser power was insufficient to maintain a uniform temperature profile across the interface, leading to inhomogeneous weld formation [36].

While the nonlinear nature of the process introduces complexity in execution, it also offers a high degree of control and adaptability. Parameters such as fluence, repetition rate, and scanning speed must be carefully balanced to avoid pre-focus absorption, beam distortion, and focal offset, all of which can degrade weld seam quality [35], [37]. To the best of the authors' knowledge, no research has thoroughly examined energy deposition in the processing area, the limiting factors across different fluence regimes and their causes, or the effect of offset on weld seam quality.

This study focuses on optimizing the welding of transparent polymers using femtosecond laser pulses, aiming to eliminate the need for absorbing elements while achieving strong bonds with homogeneous and uniform weld seams. Initially, the effects of laser parameters such as fluence per pulse, scanning speed, and repetition rate on the weld seam morphology are investigated. During this process, an offset between the intended focal position and the actual processing zone is observed. However, after pre-adjusting the laser focus to place the processing area directly at the polymer interface, non-uniformity in the weld seam becomes apparent. To investigate the optical contribution to this issue, beam-propagation simulations are conducted in Zemax OpticStudio. These simulations analyze how axial offset and lens aberrations modify the linear beam intensity distribution at the experimentally observed processing depth and how this optical distortion influences weld seam uniformity. Nonlinear propagation effects in polycarbonate are not included, as Zemax is used here solely as a linear optical diagnostic tool to assess the influence of the beam delivery optics and axial offset, rather than to model the nonlinear spatiotemporal evolution of femtosecond pulses in the material. These simulations provide insights into the beam profile at the processing site and its influence on weld seam quality. Based on these findings, we propose an optimization method that uses the highest laser fluence achievable without inducing offset. Although one pass at this level cannot create bonding, each additional pass changes the material's optical properties, increasing absorption and allowing bonding to occur after several passes without introducing offset. This strategy ensures that sufficient total energy reaches the interface, effectively eliminating the offset and producing high-quality, uniform welds with strong bond strength. Finally, the morphological, optical, chemical and mechanical properties of the samples are thoroughly analyzed to validate the effectiveness of the proposed optimization method. These evaluations offer insights into the material response under varying processing conditions, ensuring that the optimized parameters yield high-quality welds and strong mechanical performance.

2. Material and methodology

To provide an overview of the experimental workflow and the rationale behind the parameter selection and optimization steps, Fig. 1 presents a schematic summary of the methodology adopted in this study. The workflow is structured around two successive objectives. In the first stage, process parameters are systematically screened to identify a viable processing window and to distinguish between limiting regimes governed by excessive cumulative fluence or excessive fluence per pulse.

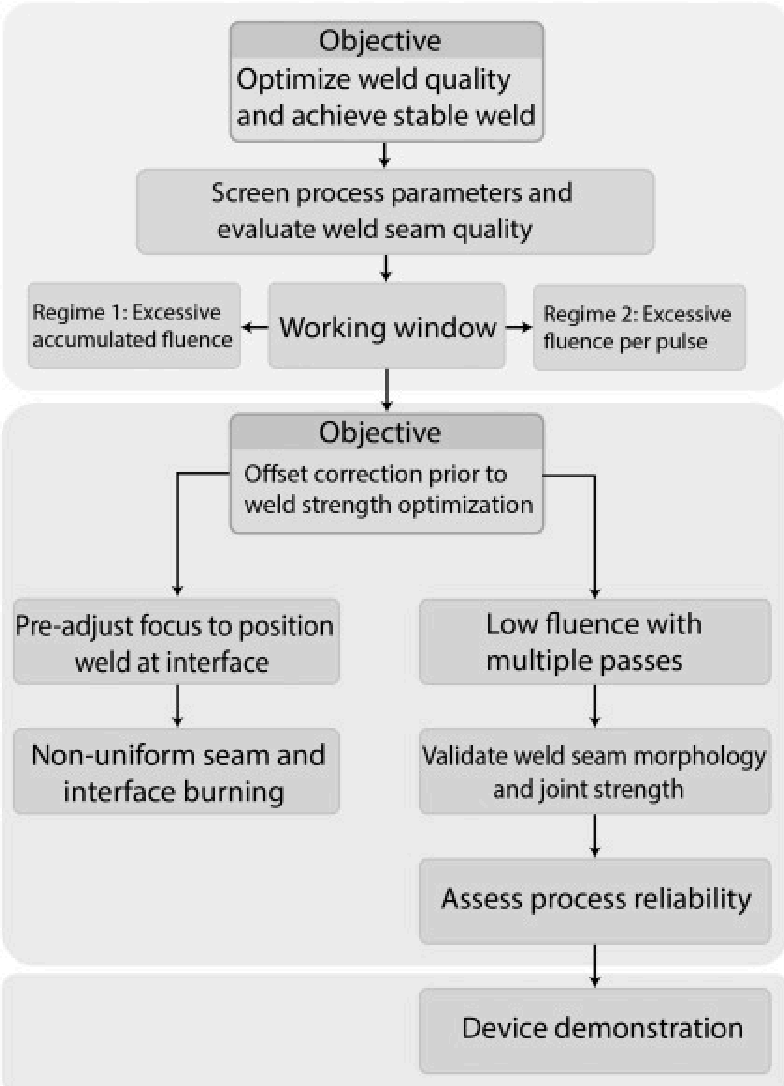

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Fig. 1. Schematic overview of the experimental workflow for femtosecond laser welding of transparent polycarbonate.

This step establishes the conditions under which homogeneous weld seams can be formed.

In the second stage, the focus shifts to understanding and correcting the weld seam offset that arises within the identified working window. The schematic illustrates that offset correction is examined before the investigation of mechanical performance; however, direct offset correction at the interface leads to non-uniform weld seams and increased interface burning compared to bulk modification. Based on this analysis, a low-fluence multi-pass strategy is developed to suppress offset formation while enabling sufficient energy accumulation at the interface. The final steps of the workflow validate the optimized process through weld seam morphology analysis, mechanical testing, reliability assessment, and device-level demonstration.

2.1. Experimental setup and characterization

In a first step, this study aims to determine the optimal working window for obtaining a high-quality weld seam. Subsequently, the bonding strength is analyzed and optimized based on the proposed methodology.

The thermoplastic material used is 1.5 mm-thick polycarbonate (PC) plates (PyraSied). PC has high stiffness (Young's modulus: 2.35 GPa), transparency, and low shrinkage, making it well-suited for point-of-care devices, such as microneedle components and microfluidic chip systems [38]. All samples are cut into plates with dimensions 75 mm × 15 mm × 1.5 mm. Although the polymer is supplied with a protective cover, it is cleaned with ethanol before processing. The spectral properties of polycarbonate (PC) are analyzed using a UV–VIS–NIR spectrometer (Agilent Cary 60 UV-VIS) to verify its transparency at the laser wavelength of 1030 nm. Effective transmission of the laser beam through the top plate is critical for directing energy to the interface between the two plates, where nonlinear absorption enables localized bonding. UV–VIS–NIR spectroscopy confirms that the plates exhibit high transmission (~90%) at 1030 nm, ensuring efficient energy delivery to the bonding region.

The optomechanical system consists of a maximum 20 W femtosecond laser source (Carbide, Light Conversion), a beam delivery path, a 2-axis optical galvanometer scanner, and an associated electronic control unit. The laser has a wavelength of 1030 ± 10 nm, a Gaussian beam profile with linear polarization, and an $M2$ factor of <1.2. The maximum repetition rate is 1 MHz, with a pulse duration of 250 fs. The beam waist is 15.0 ± 0.1 μm (radius at $1/e2$), measured using an Ophir/BeamGage beam profiler calibrated for 1030 nm, and the F-theta lens focal length is 125 mm and the entrance pupil diameter of 20 mm, corresponding to a numerical aperture of NA ≈ 0.08. Details of the studied parameters (fluence per pulse, repetition rate, and scanning speed) are summarized in Table 1. In this study, the fluence per pulse, scanning speed and repetition rate are not only practical control parameters but also directly determine the way energy is deposited at the interface. The fluence controls the peak intensity of each femtosecond pulse and therefore the onset of nonlinear absorption and plasma formation: if fluence is too low, multiphoton absorption remains below the threshold, and no material modification occurs, whereas too high fluence initiates absorption before the nominal focus, increasing the vertical offset and causing beam distortion and local damage. The scanning speed determines the spatial pulse overlap and thus the cumulative fluence delivered along the weld; low speed leads to a high number of pulses per unit area and strong heat accumulation, which can induce carbonization and bubbles, while high speed reduces pulse overlap so much that the interface does not reach the temperature required for cohesive bonding. The repetition rate sets the temporal spacing between pulses and, at fixed speed, directly scales the number of pulses per unit length: higher repetition rate enhances pulse-to-pulse heat accumulation and allows welding at lower Fluence, but also increases sensitivity to offset formation if the total deposited energy becomes too high. For this reason, these three parameters are systematically varied within the ranges of Table 1 to map both the useful welding window and the damage thresholds. The fluence values are obtained by adjusting the laser output power in 1% increments, which, after conversion, results in non-equidistant fluence steps within the reported range. Line-writing experiments are conducted to determine the optimal parameters for achieving uniform weld seam morphology by varying fluence per pulse, scanning speed, and laser repetition rate. The scanning speed is varied, starting from 5 $mm\ /s$ and then increased in steps of 10 $mm\ /s$, while the repetition rate is set at two discrete values. In practice, wider ranges of fluence per pulse and scanning speeds are investigated; however, values that caused noticeable optical damage are excluded from further investigation.

Table 1. Experimental parameters

Parameter	**Value**	**Units**
Fluence per pulse	0.06–1.78	$\mathrm{J/\ cm^2}$
Repetition rate	500 and 1000	kHz
Scanning speed	5–40	$\mathrm{mm\ /s}$

The experimental setup for bonding transparent polymers using an ultrashort pulse laser is illustrated in Fig. 2a. The laser beam is directed through a galvo scanner and focused by an F-theta lens into the interface between two transparent polymer plates mounted on an XY-stage. As shown in Fig. 2b, the plates are pressed together using a pneumatic clamping system (Festo EV-20/75), which applies a maximum force of 600 N under 0.6 MPa air pressure to ensure intimate contact during laser processing. The two 1.5 mm-thick polycarbonate plates are held inside a rigid steel frame consisting of two 4 mm-thick steel plates that fully support the polymer over the entire bonding area. This configuration effectively prevents bending of the polycarbonate during clamping, so that no additional gap is introduced by mechanical deformation of the plates. Fig. 2c illustrates that although the laser light is focused at the interface, nonlinear absorption can occur earlier within the top plate if the incoming laser pulses have sufficient intensity to exceed the threshold for multiphoton absorption. This results in a vertical offset between the focal point and the actual weld seam. This absorption behavior is shown schematically in Fig. 3a, where the absorbed area shifts upward, resulting in a wider welded region due to the Gaussian intensity profile of the focused beam. A camera aligned along the laser axis is used to measure this offset and monitor the location of the processed area relative to the initial focus. After each experiment and before disassembling the setup, weld lines are inspected using an in-system camera positioned along the beam axis. The imaging system consists of an industrial CMOS camera (2/3″ sensor format, C-mount) equipped with an Edmund Optics telecentric lens (model #67317, 8× magnification, 65 mm working distance, NA = 0.088, telecentricity 0.102°, distortion 0.14%, f/45 aperture, depth of field ±0.03 mm). This configuration enables measurement of the weld seam position relative to the interface, allowing determination of the vertical offset (z). An adjustment of the focal plane position is demonstrated by Roth et al. to be necessary under varying welding speeds to achieve effective bonding, an observation that is further supported by the offset correction approach adopted in this work [37].

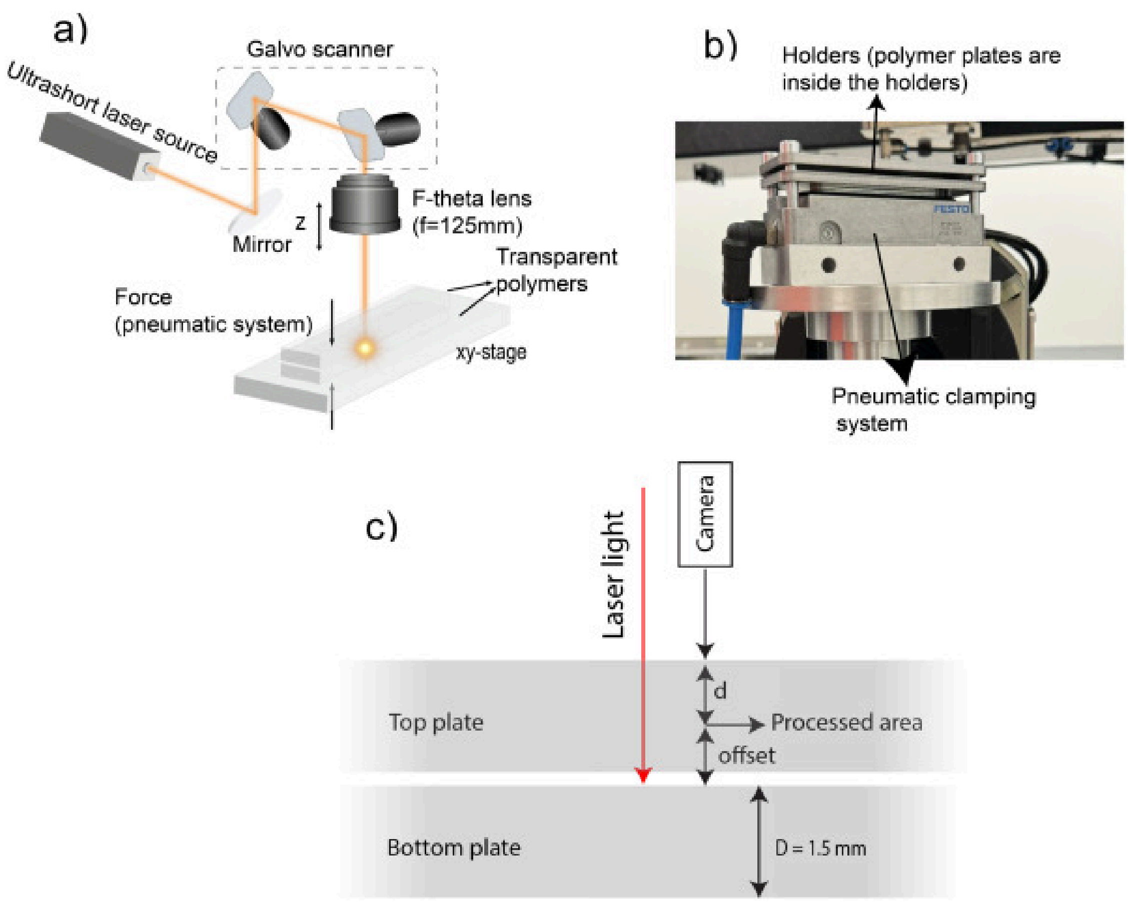

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Fig. 2. Experimental laser setup. a) Schematic overview laser setup, b) photo of actual setup and c) laser focus and weld seam offset in transparent polymer bonding

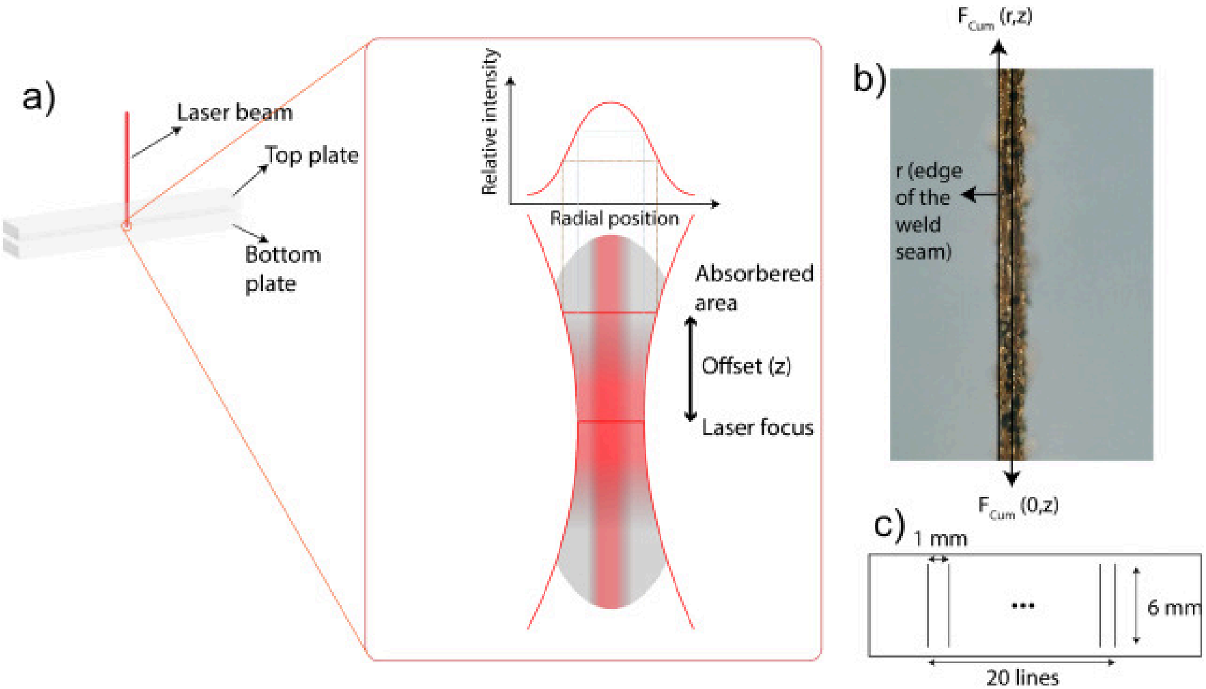

Fig. 3. a) Schematic representation of light absorption by the polycarbonate (cross-section), b) weld seam top view and c) configuration of the weld line, consisting of 20 parallel scan lines spaced by 1 mm.

The measured vertical offset is incorporated into Zemax OpticStudio (Ansys, USA) to visualize the beam profile at the weld-seam position, since the software cannot directly simulate nonlinear absorption in the polymer. Beam propagation is simulated using the sequential Huygens PSF formalism, and the model contains the complete prescription of the 125 mm F-theta lens used for beam delivery, together with the polycarbonate layers. A Gaussian input beam is defined by selecting "Entrance Pupil Diameter" as the aperture type and applying a Gaussian apodization factor of 1.0. The simulation is performed at the laser wavelength of 1030 nm under on-axis illumination. Huygens PSF calculations use 512 × 512 sampling points in both the pupil and image planes, resulting in a physical image-plane grid of 207.72 μm × 207.72 μm. Boundary handling follows the standard Kirchhoff diffraction implementation, which automatically applies zero-padding at the grid edges. Each PSF is evaluated at a single axial plane ("Image Delta = 0"). The experimentally determined offset is introduced by shifting the IMAGE-surface thickness, allowing the beam profile and intensity distribution to be reconstructed at the actual processed region. This procedure enables us to assess how the beam shape evolves at the weld seam and interpret the influence of axial shifts and optical aberrations on the resulting weld quality. This provides a more accurate representation of the beam characteristics in the processed area and helps to interpret how energy is delivered during the welding process. To assess the lateral quality of the weld, a digital microscope (Keyence VHX-6000) is used to capture images of the weld seam and measure the radial width (r) across the bonded interface, as illustrated in Fig. 3b.

This section outlines the experimental setup and procedures used to investigate bonding experiments. Each bonding sample consists of 20 parallel weld lines, spaced 1 mm apart and the number of overscans per line depends on the selected processing parameters, each 6 mm in length, designed to evaluate the mechanical strength of the joints. To facilitate lap shear strength testing, the centers of the polymer plates are intentionally offset by 10 mm along the longitudinal direction within the holders. This configuration allows each plate to be gripped by the testing apparatus during loading. The parameter values listed in Table 2 are used in the welding experiments and are selected based on the proposed methodology, which is described in detail in a subsequent section. The number of passes controls how much of that fluence is ultimately absorbed at the interface. A single pass at low fluence may leave the polymer mostly transparent, while repeated passes progressively increase absorption, allowing welding to occur without raising fluence or inducing offset. Too few passes produce weak or incomplete seams; additional passes improve bonding strength until a certain point. The morphology of the weld seams is studied for different numbers of overscans, and for the most promising weld seams, the shear strength is analyzed. Bond strength is assessed using a motorized tension and compression tester (BZ2.5/TS1S, Zwick, Germany) equipped with a 500 N load cell. Tests are conducted in displacement-controlled mode at a crosshead speed of $2\,mm\,/\,min$. For each parameter set, three replicate samples are tested. Seam width is measured individually for each sample, and the bonded area is calculated as seam width × $6\,mm$ × 20 weld lines. Shear strength is computed as the maximum force at break divided by the bonded area. This relationship is expressed as [39];

$$\tau = \frac{F}{A} \qquad (1)$$

where F represents the measured force at the break in Newtons, and τ denotes the shear strength in $N/\,mm\,2$. A is determined by multiplying the seam width, seam length, and the number of seams per sample. This approach enables a quantitative comparison of the strength of different samples. Only samples with weld material visible on both fracture surfaces are considered cohesive failures and are included in the analysis.

Table 2. Fluence per pulse requirements for varying numbers of overscans and scanning speeds

Repetition rate	**Number of passes**	**10** $\mathrm{mm/s}$	**20** $\mathrm{mm/s}$	**30** $\mathrm{mm/s}$
1 MHz	5	–	$0.15\ \mathrm{J/cm^2}$	$0.17\ \mathrm{J/cm^2}$
	7	–	$0.15\ \mathrm{J/cm^2}$	$0.17\ \mathrm{J/cm^2}$
	9	$0.08\ \mathrm{J/cm^2}$	$0.15\ \mathrm{J/cm^2}$	$0.17\ \mathrm{J/cm^2}$
	12	$0.08\ \mathrm{J/cm^2}$	–	–
	15	$0.08\ \mathrm{J/cm^2}$	–	–

FTIR analysis is conducted using a Perkin Elmer Spectrum 65 FT-IR spectrometer in ATR mode over the range 600–4000 cm^{-1}. All spectra are normalized to the aromatic $C = C$ stretching band at 1500–1515 cm^{-1} for comparative analysis.

To demonstrate the practical application of the proposed femtosecond laser welding technique, a microfluidic device is fabricated by bonding an injection-molded polycarbonate substrate containing microchannels to a flat polycarbonate cover plate (Fig. 4). The device consists of a straight microchannel measuring 22 mm in length, 500 μm in width, and approximately 500 μm in depth. Circular inlet and outlet reservoirs are integrated for fluidic interfacing. The processed welding area surrounds the microchannel, with a minimum distance of 0.2 mm between the channel wall and the welded region.

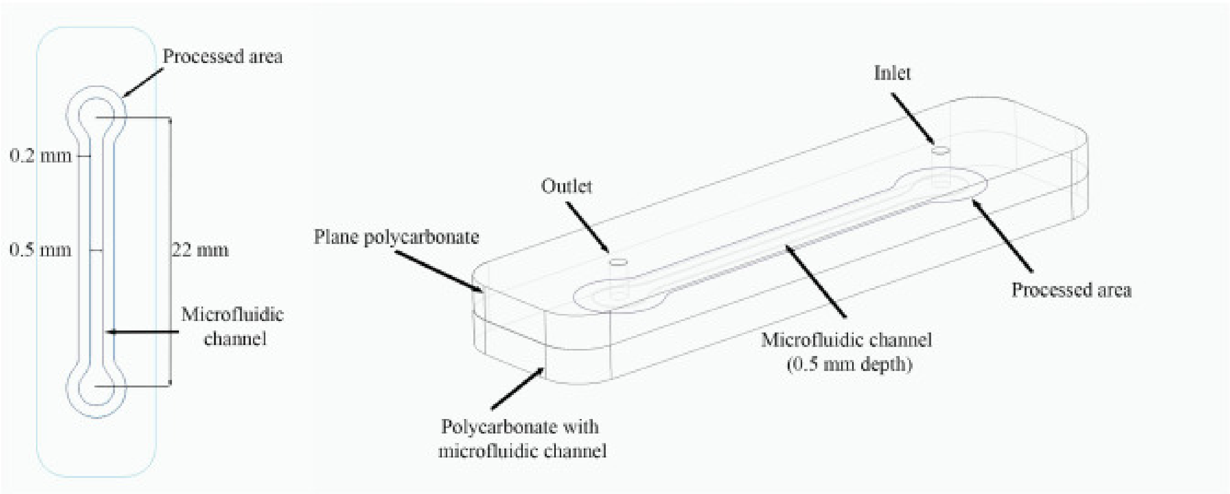

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Fig. 4. Polycarbonate microfluidic device with a 22 mm × 500 $\mu\mathrm{m}$ × 500 $\mu\mathrm{m}$ channel, bonded to a flat cover plate. The processed welding area surrounds the channel with 0.2 mm spacing.

To evaluate both the leakage performance and burst pressure of the laser-welded polycarbonate microfluidic devices, a pressure controller (LineUp Flow EZ™ Series 1000 $mBar$, Fluigent, France) is used to flow dyed distilled water into the microchannels through standard fluidic connectors. The same setup is used for both tests. For the leakage test, the outlet of the device remains open to allow continuous flow, while the system is pressurized to 1 bar, which corresponds to a flow rate of 120 $\mu L\ min^{-1}$, and the device is visually inspected for any signs of fluid leakage around the welded regions.

2.2. Theoretical calculations

Cumulative fluence (F_{Cum}), energy dose, is crucial to understanding the bonding mechanism of transparent polymers, as it reflects the effects of multiple pulses instead of focusing solely on individual pulse fluence. The heat accumulation, structural changes, and chemical reactions in the material depend on the total energy delivered over time, not just the energy from a single pulse [40]. Considering the Gaussian profile, the cumulative fluence of the weld seam is calculated at both the center and the edges as they reflect the processed area with maximum and minimum delivered energy areas, respectively. Fig. 3 shows the schematic of light absorption in polycarbonate and the top view of the weld seam. The offset (z) is measured using the telecentric camera, and the weld width is obtained from top-view microscopy. For a Gaussian beam, the cumulative fluence can be calculated using the fluence $F(r, z)$ at a radial distance r from the beam axis and position z and the effective number of pulses per unit area as follows [41]:

$$F_{Cum}(r, z) = F(r, z) \times n_{eff} \tag{2}$$

$$F(r, z) = F_0 \left(\frac{\omega_0}{\omega(z)} \right)^2 \exp\left(-2 \frac{r^2}{\omega(z)^2} \right) \tag{3}$$

$$F_0 = \frac{2E}{\pi \omega_0^2} \tag{4}$$

$$\omega(z)= \omega_0\sqrt{1+\left(\frac{z}{z_R}\right)^2} \tag{5}$$

$$z_R = \frac{\pi\omega_0^2}{\lambda} \tag{6}$$

where F_0 is the peak fluence at the focus ($z = 0$), ω_0 is the beam waist (radius), $\omega(z)$ is the beam radius at distance z, E *is the pulse energy,* Z_R is Rayleigh's range and λ is the laser source wavelength.

The cumulative fluence reported in this work represents a delivered areal dose evaluated at the onset depth of visible modification (top of the bulk-modified region), rather than a direct measure of absorbed volumetric energy density $\left(J/\,cm^3\right)$. Energy deposition during femtosecond laser processing can extend over a significant axial length, and the present approach does not quantify how much of the incident energy is absorbed above, within, or below this reference plane. In addition, the modified region exhibits a strongly depth-dependent width, making any reconstruction of volumetric energy density highly uncertain. For these reasons, cumulative fluence is used as a comparative metric to identify processing regimes and trends in bonding behavior.

n_{eff} is the effective number of pulses per unit area, defined as:

$$n_{eff} = \frac{1}{1 - OL} \tag{7}$$

where OL is the pulse overlap, defined as:

$$OL = \frac{2\times r^2\times cos^{-1}\left(\frac{\Delta x}{2r}\right) - \frac{\Delta x}{r}\times\sqrt{4\times r^2\times \Delta x^2}}{\pi\times r^2} \tag{8}$$

where Δx is the spatial distance between two consecutive laser pulses and is defined as $\Delta x = scanning\ speed/\ repetitionrate.$

After analyzing the lap-shear data, for each scanning speed (10, 20, and 30 $mm\,/s$), the parameter set corresponding to the highest fluence that does not induce an offset, combined with using at least 9 overscans for 10 $mm\,/s$ and 9 overscans for 20 and 30

mm/s is selected for a two-parameter Weibull analysis to quantify the characteristic strength and variability of the welds. The cumulative probability of failure is expressed as:

$$F(\sigma) = 1 - exp\left[-\left(\frac{\sigma}{\sigma_0}\right)^m\right] \quad (9)$$

where β (Weibull modulus) describes the scatter in strength data, higher values indicate more consistent joints, and σ_0 (scale parameter) represents the characteristic strength at which 63.2% of specimens are expected to fail. Weibull analysis is important as it quantifies weld reliability through the Weibull modulus and determines the characteristic strength.

3. Results and discussion

3.1. Optical response of polycarbonate

Fig. 5 shows the UV–Vis-NIR transmission spectrum of PC. This spectrum provides insight into the optical behavior of the material across a broad wavelength range.

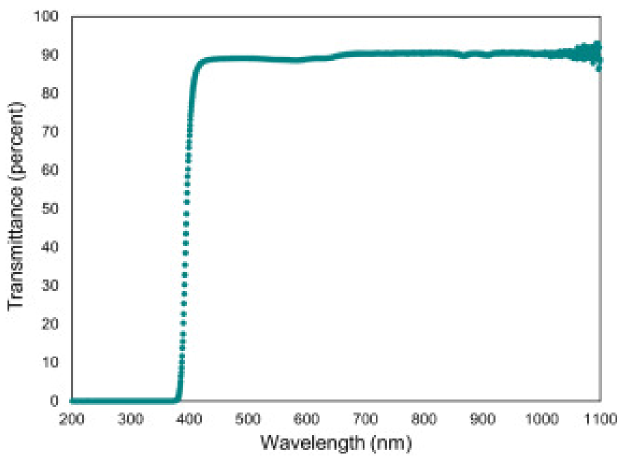

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Fig. 5. Transmittance spectrum from UV–Vis-NIR spectroscopy of polycarbonate.

The graph indicates that PC is approximately 90% transparent at the working wavelength of 1030 nm laser. PC begins to absorb light linearly around 390 nm, corresponding to a bandgap of 3.18 eV. Given the photon energy of 1.203 eV, at least 3 photons are required to excite electrons from the valence band to the conduction band via multiphoton absorption. The combination of high transparency, which allows light to pass through the volume, and the potential for non-linear absorption of femtosecond laser pulses enables precise micromachining within the bulk or at the interface of the polymer.

3.2. Effect of fluence and scanning speed on weld formation

Fig. 6 presents a cross-sectional view of a transparent polycarbonate sample welded using a femtosecond laser operating at a 1 MHz repetition rate and a scanning speed of 20 mm/s with a single overscan. This parameter set is shown as a representative example because it clearly illustrates the progression of the weld seam with increasing fluence per pulse, which in this case rises from 0.17 to 0.66 J/cm^2 from right to left. A single overscan is used for this example.

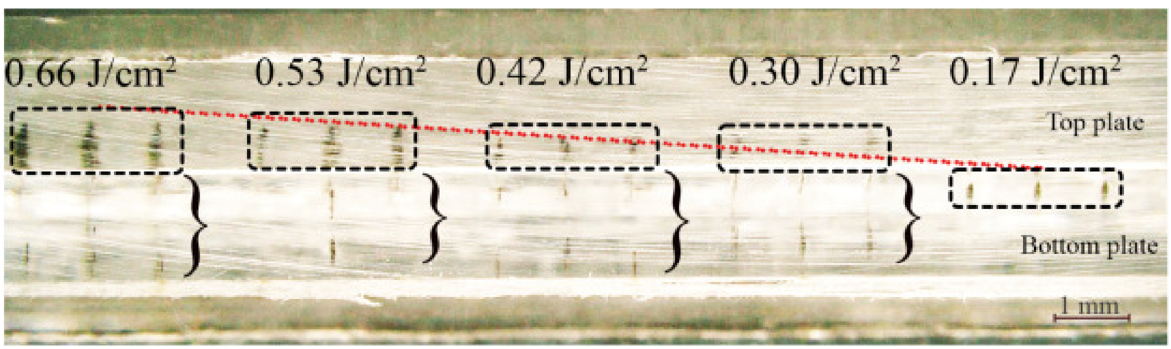

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Fig. 6. Cross-sectional microscopy images of transparent polymer welds produced using a femtosecond laser at 1 MHz repetition rate with increasing fluence per pulse.

Rectangular boxes mark the main welding zones with highest energy deposition, curly brackets indicate laser-induced effects at different depths, and the red dotted line highlights the upward shift in the weld seam position with increasing fluence.

Rectangular boxes outline the main welding zones. These zones represent the regions with the highest laser energy deposition and material modification. Curly brackets below each weld highlight laser-induced effects observed at varying depths within the material. Due to the transparency of the polymer, these effects are all visible in the same 2D cross-sectional image as the main processed area, but they do not necessarily share the same X-Y position. The inconsistency in energy deposition along the depth is likely caused by a distorted energy distribution, potentially resulting from early absorption or nonlinear phenomena such as self-focusing, filamentation, or plasma formation [42]. Such distortion leads to variations in bonding quality and makes it difficult to precisely control the location of processing area. This behavior highlights a trade-off: higher fluences provide the necessary energy for potential bonding but introduce depth-dependent absorption inconsistencies. In contrast, lower fluences offer more uniform in-depth interaction but may fail to form bonding. Therefore, identifying an optimal processing window requires balancing sufficient energy input with controlled spatial confinement of absorption.

A clear upward shift in the weld seam position is observed with increasing fluence, as shown by the red dotted line. This indicates that at higher fluence levels, absorption begins earlier along the optical axis. As fluence increases, stronger interactions occur closer to the surface, resulting in shallower process locations. Due to the difficulty in clearly distinguishing weld seams and energy deposition depths from side-view images, evaluations of weld quality will be based on top-view images. It will be assumed that the primary absorption depth corresponds to the visible top of the weld seam, where nonlinear absorption likely reaches the required threshold for bonding. While this assumption does not fully capture the subsurface behavior, it provides a consistent and practical reference point for comparative analysis.

The effective number of pulses delivered to a point on the interface is defined in Eq. (7) and is directly determined by pulse overlap. In Fig. 7, the dependence of the effective number of pulses on fluence per pulse is presented for all parameter sets. To illustrate this behavior, the effective number of pulses is compared for scanning speeds of 5 mm/s and 10 mm/s, both with a repetition rate of 500 kHz. At a representative fluence of 0.08 J/cm^2, the number of pulses accumulated at 5 mm/s (2710 pulses) is found to exceed that at 10 mm/s (1285 pulses) by approximately a factor of two, reflecting the substantially higher spatial overlap at the lower speed. The same trend is observed at 1 MHz, where low-speed processing again results in several times more pulses per point than high-speed processing. Conversely, when the scanning speed is held constant (e.g., 20 mm/s at 0.65 J/cm^2), an increase in repetition rate from 500 kHz to 1 MHz nearly doubles the effective number of pulses, increasing from 1020 to 2509 pulses, consistent with the temporal pulse spacing. These two comparisons show that both parameters strongly modulate pulse accumulation. Since cumulative fluence is given by the product of the fluence per pulse and the effective number of pulses, these trends demonstrate how scanning speed and repetition rate jointly determine the cumulative energy delivered to the interface.

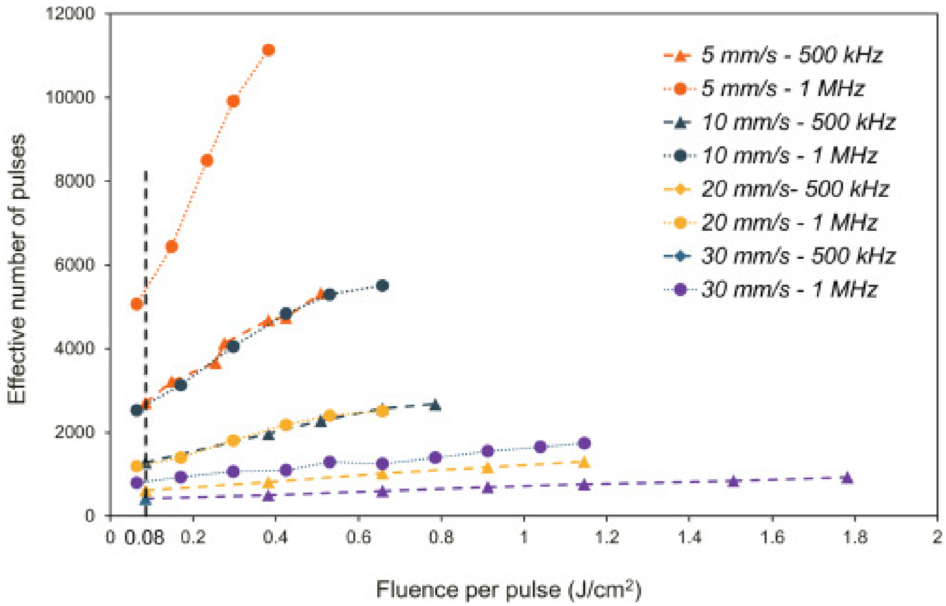

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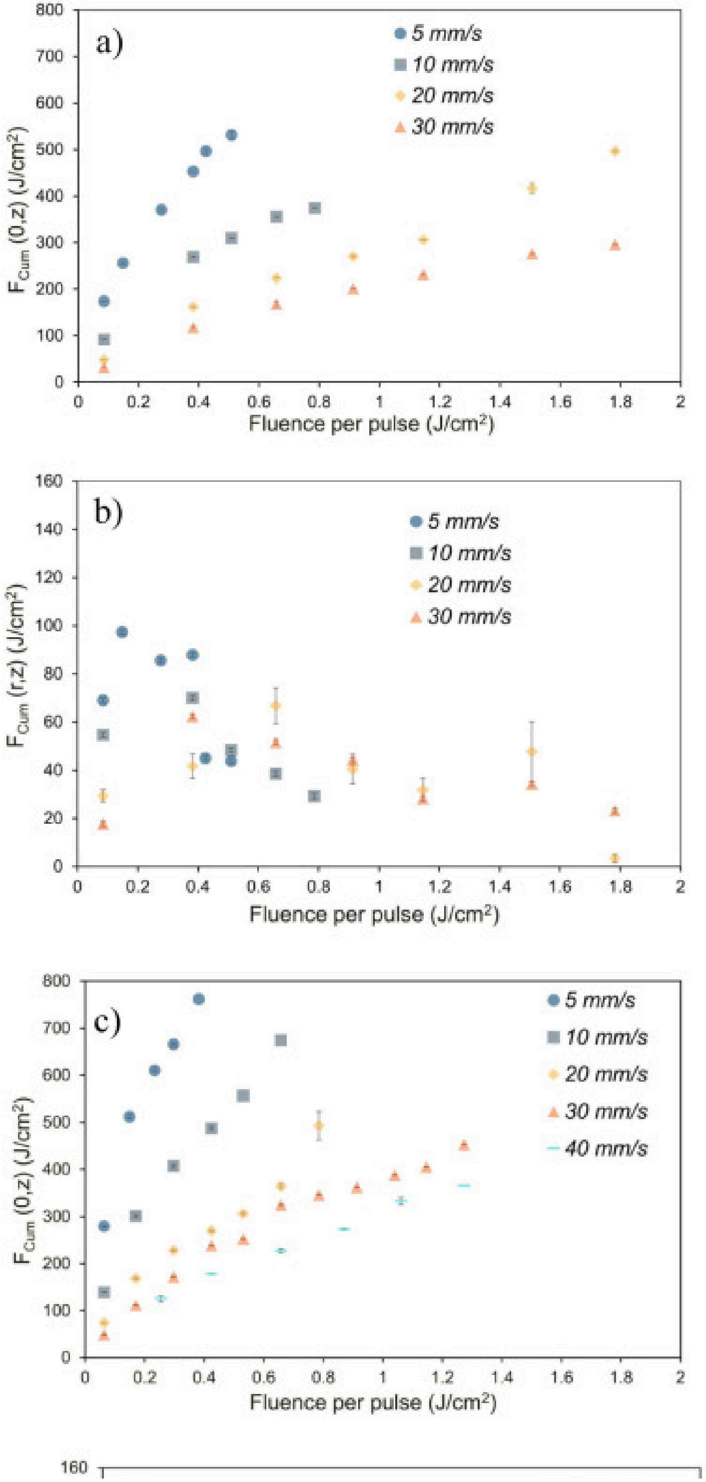
Fig. 7. Effective number of pulses as a function of fluence per pulse for different scanning speeds and repetition rates.

To evaluate the energy distribution within the bonded region, the cumulative fluence is calculated at both the center and the edges of the weld seams. The total energy per unit area is evaluated at the top of the bulk modified area by applying (i) z, the vertical offset from the focal plane measured via the internal camera, and (ii) r, defined as half the weld-seam width, into Eq. (2). The fluence at two key points, the edges and the center of the weld, is particularly interesting. The center value represents the maximum fluence delivered to the area, as the radial distance (r) is zero, while the minimum energy is delivered to the edges.

Fig. 8 illustrates the cumulative fluence: (a, c) at the center and (b, d) at the edge of the weld seam, corresponding to locations receiving maximum and minimum energy, respectively. The relationship between cumulative fluence $F_{Cum}(0, z)$ (J/cm^2), and fluence per pulse (J/cm^2) at different scanning speeds are shown in Fig. 8a for 500 kHz repetition rate and in Fig. 8c for 1 MHz. Fig. 8b and d shows the corresponding cumulative fluence $F_{Cum}(r, z)$ J/cm^2 for 500 kHz and 1 MHz, respectively.


a)
5 mm/s
10 mm/s
20 mm/s
30 mm/s
F_Cum (0,z) (J/cm²)
Fluence per pulse (J/cm²)
b)
5 mm/s
10 mm/s
20 mm/s
30 mm/s
F_Cum (r,z) (J/cm²)
Fluence per pulse (J/cm²)
c)
5 mm/s
10 mm/s
20 mm/s
30 mm/s
40 mm/s
F_Cum (0,z) (J/cm²)
Fluence per pulse (J/cm²)
160

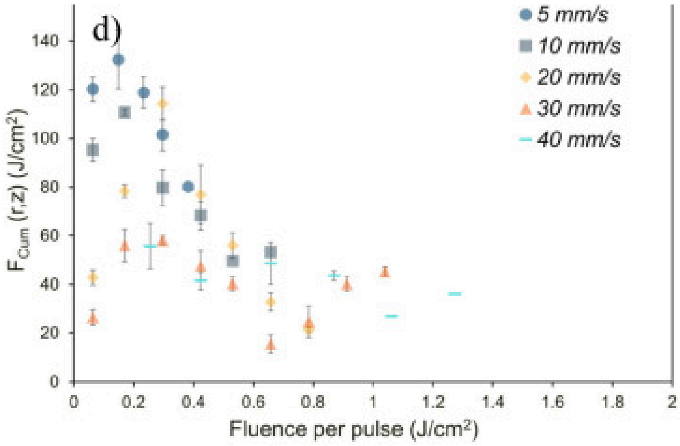


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Fig. 8. The cumulated fluence of weld seam for 500 kHz at (a) the center and b) the edge; and for 1 MHz at (c) the center and d) the edge. The error bars represent the standard deviations.

As shown in Fig. 8a and c, the cumulative fluence reaches higher values at a repetition rate of 1 MHz compared to 500 kHz, given the same fluence per pulse and scanning speed [43]. This increase is attributed to the higher pulse overlap per unit area at the higher repetition rate. For both repetition rates, the cumulative fluence at the center of the weld seam increases with fluence per pulse across all scanning speeds. The range of fluence per pulse investigated varies with scanning speed, as scanning speed directly influences pulse overlap and, consequently, cumulative fluence. At lower scanning speeds, the increased overlap leads to higher cumulative fluence, which can result in early thermal damage to the polymer. Under these conditions, the fluence per pulse must be reduced to prevent material degradation. Conversely, increasing the scanning speed reduces cumulative fluence due to fewer overlapping pulses. As a result, higher fluence per pulse is required at faster scanning speeds to maintain sufficient energy deposition for effective bonding. However, exceeding a certain fluence per pulse threshold can lead to localized damage or burning of the polycarbonate due to excessive pulse intensity. For example, experimental observations show that at a repetition rate of 1 MHz and a scanning speed of 5 $mm\,/s$, the maximum fluence per pulse that can be applied without causing burning is approximately 0.38 $J/\,cm^2$, corresponding to a cumulative fluence of 761.8 $J/\,cm^2$. At higher scanning speeds of 10, 20, and 30 $mm\,/s$, the corresponding maximum cumulative fluences before the onset of surface degradation on the front surface of the top plate are 674.5, 492.9, and 405.5 $J/\,cm^2$, respectively. These values correspond to fluence per pulse levels of 0.65, 0.78, and 1.14 $J/\,cm^2$. At 40 $mm\,/s$, bonding remains achievable at a fluence per pulse of 1.27 $J/\,cm^2$, resulting in a cumulative fluence of 365.5 $J/\,cm^2$. These findings indicate that the dominant factor contributing to damage varies with scanning speed: at lower speeds, damage is primarily due to high cumulative fluence, while at higher speeds, it results from elevated fluence per pulse. Therefore, cumulative fluence alone is not a sufficient predictor of damage. Instead, a careful balance between fluence per pulse and cumulative fluence is required to ensure adequate nonlinear absorption for bonding while avoiding thermal damage.

Fig. 8b and d illustrates the relationship between the cumulative fluence at the edge of the weld seam, $F_{Cum}(r,z)$ $(J/\,cm^2)$, and the fluence per pulse $(J/\,cm^2)$ at various scanning speeds for 500 kHz and 1 MHz, respectively. As previously explained, the cumulative fluence generally decreases with increasing scanning speed due to reduced pulse overlap. Notably, a counterintuitive behavior is observed wherein cumulative fluence exhibits a decreasing trend with increasing fluence per pulse. At lower fluence per pulse, laser pulses pass through the polymer and concentrate at the focal point, resulting in a smaller beam diameter. This confines the energy distribution to a tighter region, leading to higher energy density at the weld seam, including the edges of the weld seam. In contrast, at higher fluence per pulse, the incoming light possesses sufficient energy to trigger nonlinear absorption before the beam reaches the nominal focus. As previously discussed, this causes energy deposition to begin earlier along the beam path, shifting in the effective processing location, referred to as the offset, resulting in a broader weld seam. This earlier onset of absorption spreads the energy over a larger volume, thereby reducing the cumulative fluence at the edges, considering the Gaussian beam profile.

Consequently, lower fluence per pulse levels tend to produce higher and more concentrated cumulative fluence, which is beneficial for achieving stronger and more predictable bonds.

Fig. 9 illustrates the ratio of cumulative fluence at a radial position $F_{Cum}(r,z)$ to that at the center $F_{Cum}(0,z)$ as a function of fluence per pulse, shown for two repetition rates: (a) 500 kHz and (b) 1 MHz. A higher ratio indicates a more uniform energy distribution across the weld seam, meaning the center and edges receive similar fluences, an ideal condition for consistent weld quality. As previously discussed, increasing the energy density shifts the weld seam to shift upward and broadens it. While the cumulative fluence at the center may increase or stabilize with higher fluence per pulse, the fluence at the edges decreases. This results in a growing gap between center and edge fluences, leading to an increase in inhomogeneity in energy distribution. Such non-uniformity can negatively impact weld quality by making the process less stable and predictable. This trend is reflected in both figures, where all scanning speeds show a decreasing ratio with increasing fluence per pulse. Notably, the highest ratios for all scanning speeds occur at the lower end of the fluence range in both repetition rates, reinforcing the conclusion that lower fluence per pulse generally yields better uniformity in energy distribution.

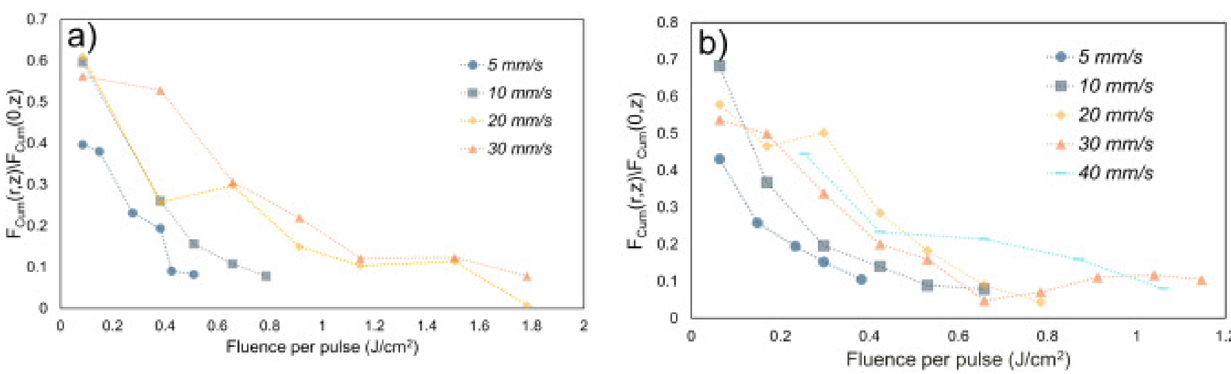

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Fig. 9. Ratio of cumulative fluence at radial position $F_{Cum}(r, z)$ to that at the center $F_{Cum}(0, z)$ as a function of fluence per pulse for various scanning speeds. With a repetition rate of (a) 500 kHz and (b) 1 MHz.

Fig. 10 presents microscopic images of the top-view of the weld seams at various cumulative fluences and speed levels, processed using a 500 kHz repetition rate.

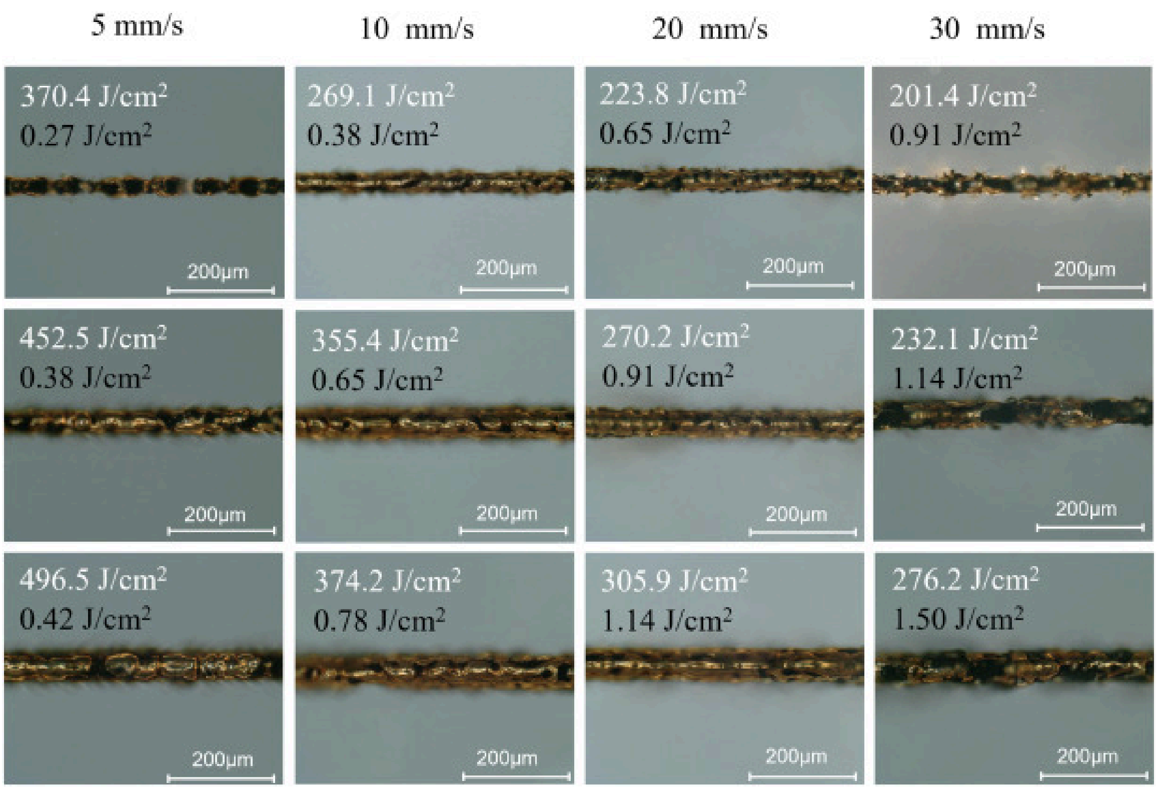

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Fig. 10. 3D digital images of weld seams at different scanning speeds and pulse energies at 500 kHz. The white font indicates the cumulative fluence, while the black font represents the fluence per pulse.

As seen in the images of the 5 mm/s column, the weld seams contain black spots, indicating localized burning of the polycarbonate caused by higher cumulative fluence values, which is an undesirable outcome. In contrast, the weld seams at scanning speeds of 10 mm/s and 20 mm/s show greater homogeneity and fewer black spots. This observation suggests that lower accumulated fluence at higher scanning speeds leads to reduced thermal damage. To ensure effective bonding despite the reduced pulse overlap at high scanning speeds, higher fluence per pulse values are necessary. More specifically, at a scanning speed of 30 mm/s, the fluence per pulse levels are generally high, increasing the risk of burning the polycarbonate, while at lower energy levels, effective bonding does not occur [44]. As already indicated, polycarbonate can be damaged either by excessive cumulative fluence at low scanning speeds or by high fluence per pulse at high scanning speeds. At a repetition rate of 500 kHz, the optimal morphological bonding range, assessed by visual inspection of the weld seam, generally lies between 10 and 20 mm/s, corresponding to fluence per pulse values of 0.38 – 0.78 J/cm^2 and 0.65 – 1.14 J/cm^2, respectively.

Fig. 11 shows microscopic images of samples processed at a repetition rate of 1 MHz. At this frequency, the increased pulse overlap results in higher cumulative fluence, requiring lower fluence per pulse. Consequently, the working window at 1 MHz is broader than that at 500 kHz, as the reduced fluence per pulse delays the onset of damage, typically caused by excessive energy input per pulse. This allows for the use of higher scanning speeds at 1 MHz, which is a significant advantage. Following the same rationale as in the 500 kHz case, the optimal processing window, determined visually by weld seam inspection, greater homogeneity and fewer black spots corresponds to scanning speeds of 10, 20, and 30 mm/s. These speeds are associated with fluence per pulse ranges of 0.29–0.53 J/cm^2, 0.42–0.65 J/cm^2, and 0.42–0.65 J/cm^2, respectively. At lower scanning speeds, excessive cumulative fluence can cause material damage, while at higher speeds, high fluence per pulse may locally degrade the weld quality.

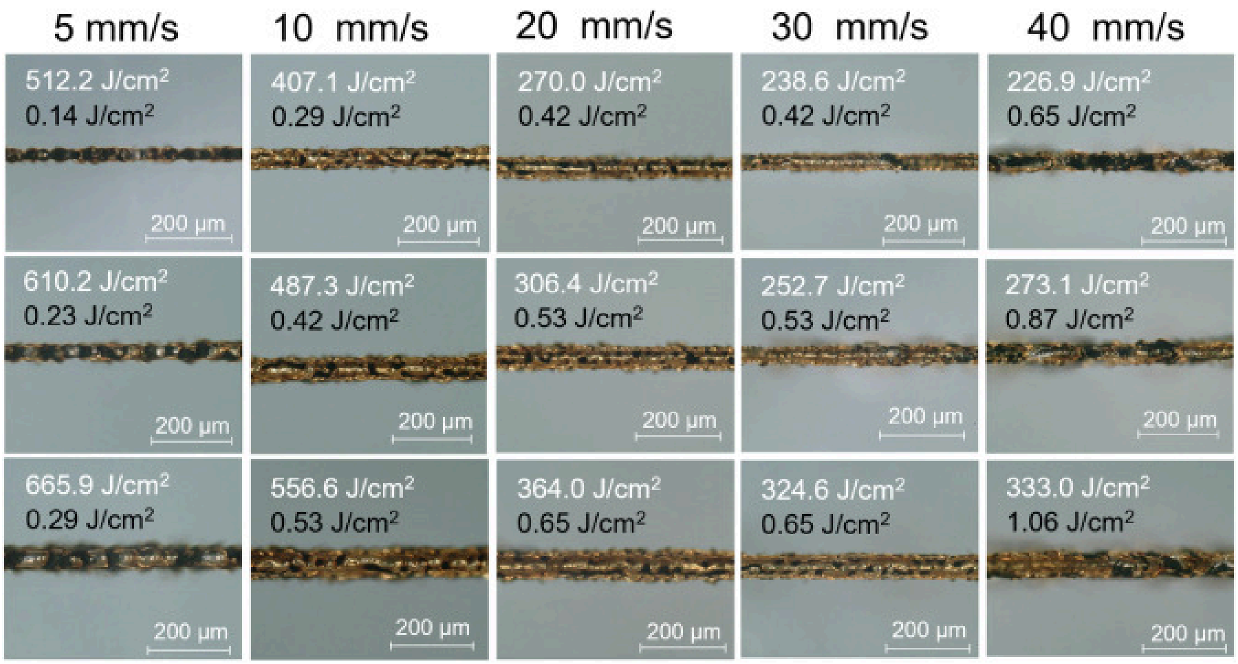

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Fig. 11. 3D digital images of weld seams at different scanning speeds and pulse energies at 1 MHz. The white font indicates the cumulative fluence, while the black font represents the fluence per pulse.

3.3. Offset measurement and beam distortion analysis

Fig. 12 shows the relationship between the measured offset, defined as the distance between the processed area and the focal plane, and fluence per pulse for different scanning speeds and repetition rates.

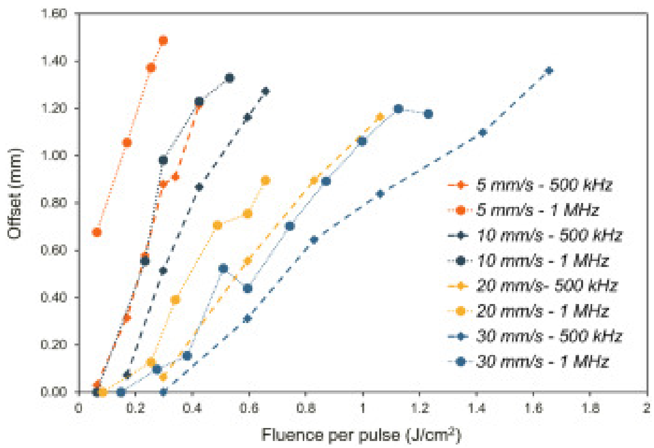

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Fig. 12. Weld seam offset vs. fluence per pulse for different scanning speeds and repetition rates.

A lower offset generally indicates a better process, as it suggests that energy absorption occurs closer to the focal plane, at the interface between the two polymer plates, where the beam profile is closest to the ideal shape of the laser source. In general, the offset increases with fluence; however, the rate of increase is influenced by other processing parameters, such as repetition rate and scanning speed. A higher repetition rate of 1 MHz consistently results in a greater offset compared to 500 kHz, which can be attributed to a higher effective number of pulses per unit area, leading to increased thermal accumulation. At lower scanning speeds, such as 5 mm/s and 10 mm/s, the offset rises sharply even at low fluence per pulse due to thermal accumulation. In contrast, at higher speeds like 20 mm/s and 30 mm/s, the increase is more gradual.

Table 3, summarizes the working windows of fluence per pulse and the corresponding weld seam offsets for different scanning speeds and repetition rates. Although offset values are generally lower at 500 kHz, this does not necessarily indicate better welding performance. As shown in the table, the optimal fluence per pulse range for 1 MHz is lower than that for 500 kHz. This lower energy input per pulse leads to lower offset values. When comparing the optimal working windows of both repetition rates, it becomes evident that 1 MHz achieves lower offset values within its optimal range. This conclusion is consistent with previous studies, which have also highlighted the advantages of using higher repetition rates in similar applications [45], [46].

Table 3. Working window of fluence per pulse and corresponding weld seam offset for different scanning speeds and repetition rates

	500 kHz		**1** MHz		
	10 mm /s	**20** mm /s	**10** mm /s	**20** mm /s	**30** mm /s
Fluence per pulse (J/cm^2)	0.38–0.78	0.65–1.14	0.29–0.53	0.42–0.65	0.42–0.65
Offset (mm)	0.16–0.92	0.50–0.90	0.1–0.81	0.13–0.52	0.10–0.33

Once the local energy density surpasses the nonlinear absorption threshold of polycarbonate, nonlinear absorption is initiated. The deposited energy causes a local temperature rise. At elevated repetition rates or low scanning speeds, this heat does not dissipate before the next pulse arrives, leading to thermal accumulation. As the temperature increases, the material becomes more absorbing, so an even larger fraction of the incoming energy is absorbed earlier along the beam path. This positive feedback shifts the effective absorption zone upward, contributing to the increase in offset.

After determining the offset, the focal position is pre-adjusted to ensure that the laser beam is absorbed at the intended polymer–polymer interface. This adjustment shifts the processing region from within the bulk material to the interface between the polymer plates, where new phenomena are observed. Fig. 13 presents 3D digital images of weld seams obtained under pre-adjusted focusing conditions, targeting the optimized interface region. The processing is performed at a scanning speed of 30 mm/s and a repetition rate of 1 MHz. Despite the focal pre-adjustment, the weld lines show noticeable variation: more burned points and some regions appear sharp, while others are faint. One observed outcome is increased burning, likely due to the presence of oxygen at the interface, which enhances oxidative reactions. The availability of oxygen can facilitate exothermic decomposition pathways, leading to intensified localized heating and material degradation [47]. Also, inconsistencies in weld seam morphology become more pronounced at the interface, with variations in sharpness suggesting uneven processing depths and inhomogeneous energy deposition. As highlighted in the white and green annotated regions of Fig. 13, these differences arise because different parts of the seam lie at slightly different axial positions within the interface. The white-marked regions correspond to areas that fall within the microscope's focal plane, resulting in sharp and clearly defined morphology. In contrast, the green-marked regions fall outside the focal plane, causing these features to appear softer or partially blurred. This behavior confirms that the weld seam does not lie at a single uniform depth; instead, different portions of the seam occupy different axial positions, which directly leads to the observed contrast between the in-focus (white) and out-of-focus (green) regions. When the laser pulse is absorbed within the bulk polymer, the continuous lattice confines most of the energy near the processing zone. In this case, the main processed area, which is visible in top-view microscopy, is well defined and corresponds to the central region of highest energy deposition. Although small fluctuations in the beam profile or system stability may cause some of the laser energy to pass through and be absorbed at other depths, these out-of-focus absorption regions are generally not visible in top-view images. However, cross-sectional observations, as shown in Fig. 6, reveal additional absorption zones outside the main weld seam. These are likely influenced by the beam profile and energy distribution, which will be discussed in the following section. At the interface, however, the situation becomes more complex. In addition to the influence of the beam profile, the presence of a physical gap between the plates increases sensitivity to fluctuations. Since the light must still be absorbed within the polymer, and the separation between the plates is significantly larger than the molecular spacing in the bulk, even small variations in the beam or setup can lead to absorption at widely varying depths and less uniform weld seams. This increased sensitivity to both beam profile and depth variation at the interface underscores the importance of considering the spatial intensity distribution of the laser. The role of beam shape, particularly under slightly out-of-focus conditions, is examined in the next section. Given these observations, a deeper understanding of how optical focusing affects energy deposition is essential. Precision in laser beam focusing is critical to achieve high-quality welds, especially in ultrashort-pulse polymer bonding, where even minor optical aberrations can drastically affect performance. The focusing optics of the laser system consist of an F-theta lens, designed to deliver a Gaussian beam at the focal plane. To evaluate the impact of offset on beam quality and consequently on weld seam quality, the existing F-theta lens is simulated using Zemax, a powerful optical design tool for analyzing propagation and aberrations.

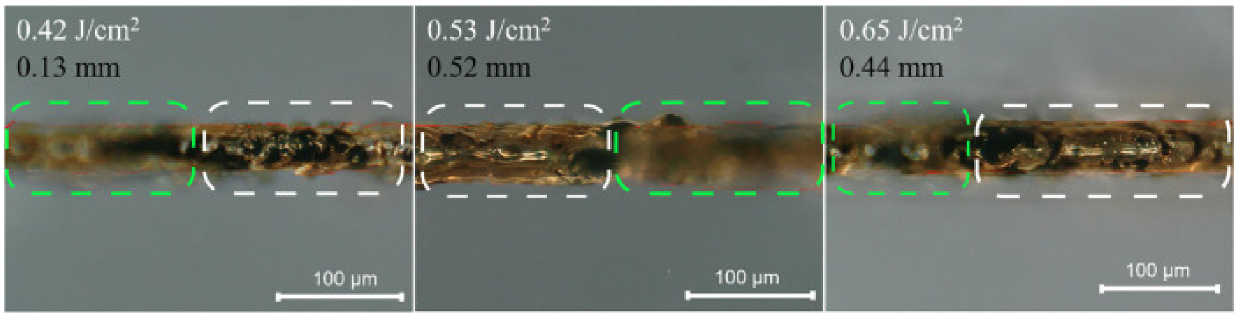

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Fig. 13. 3D digital images of the weld seam obtained under pre-adjusted focusing conditions, targeting the optimized interface region. Processing was performed at a scanning speed of 30 mm/s and a repetition rate of 1 MHz. White text indicates the fluence per pules and black shows corresponding adjusted values. The white dashed regions indicate portions of the weld seam that lie within the microscope focal plane and therefore appear sharp and well-defined, whereas the green dashed regions correspond to areas located outside the focal plane, resulting in a softer or partially blurred appearance.

As shown in Fig. 14, the Point Spread Function (PSF) is used to assess beam quality at various offsets. The PSF defines the irradiance dispersion resulting from a single-point source and shows how an optical system responds to it. Since the laser system's configuration remains fixed throughout the experiments, the observed variations in beam quality at different offsets can be attributed to field-dependent optical aberrations such as spherical distortion and defocus, introduced by the F-theta lens as the beam moves off-axis.

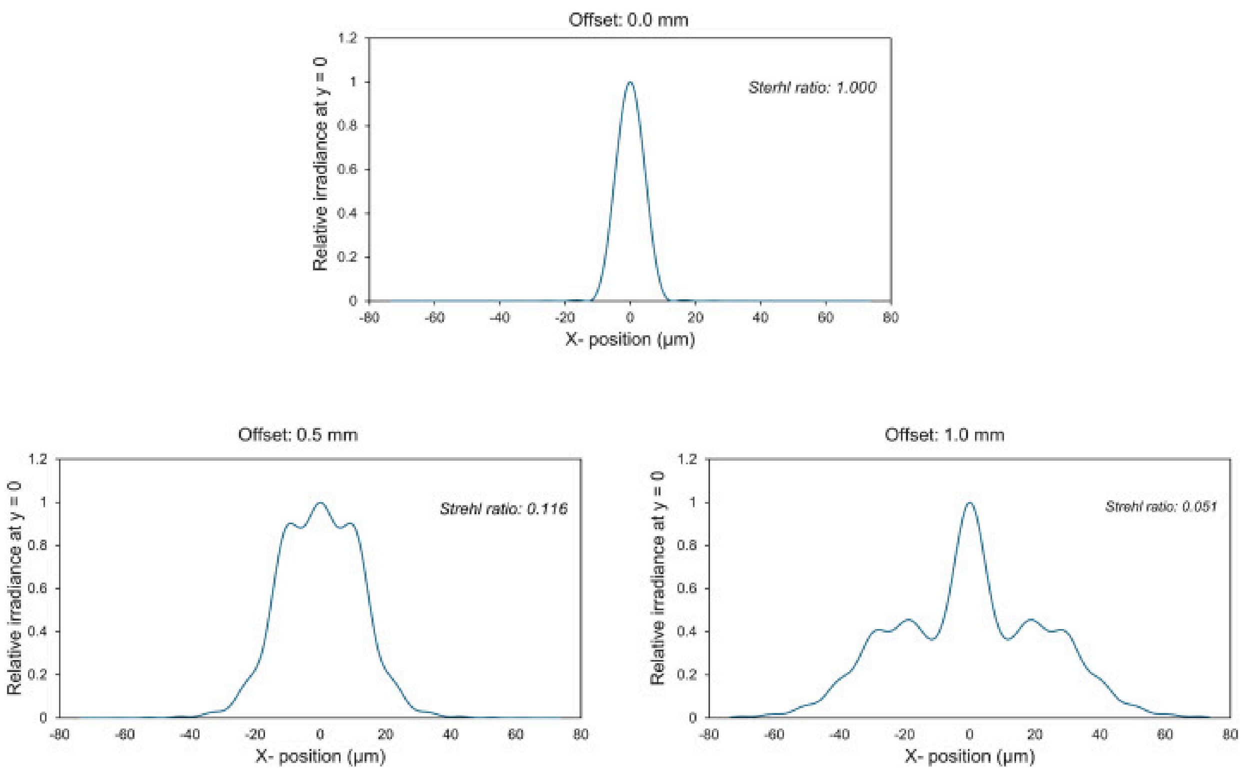

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Fig. 14. Comparison of Huygens PSF cross-sections at different offsets at 0, 0.5 and 1 mm.

At the focal point (0 mm), the beam maintains its Gaussian profile, ensuring optimal processing quality and predictability. However, as the offset increases beyond 0.5 mm, the beam profile becomes increasingly distorted, exhibiting side lobes and intensity fluctuations. These distortions are particularly critical in ultrashort-pulse polymer bonding, where nonlinear absorption is highly sensitive to local intensity distribution. Although the fluence per pulse required to initiate bonding is relatively low, the extremely high peak power of ultrashort pulses makes the process highly susceptible to spatial intensity variations. As a result, absorption may occur at varying depths within the polymer depending on the offset.

To quantitatively compare beam quality at different offsets, the Strehl ratio is used. This metric compares the peak intensity of the beam to that of an ideal, aberration-free system. A Strehl ratio of 1 indicates perfect optical performance, while lower values reflect increasing aberrations. For offsets of 0, 0.5, and 1 mm, the Strehl ratios are 1, 0.116, and 0.051, respectively, demonstrating a steep decline in beam quality as the offset increases. Thus, the variation in beam profile at different offsets leads to inconsistent energy distribution at the processed area, resulting in noticeable differences in weld seam quality and reducing both the precision and predictability of the laser bonding process. These results highlight that while geometric offset compensation is essential, it may not be sufficient to ensure weld homogeneity at the material interface.

Although parameter optimization aims to improve weld seam quality, it introduces challenges related to offset-induced optical aberrations, resulting in weld inconsistencies. Attempts to solve this by pre-adjusting for the offset have proven ineffective, highlighting the need for a more robust solution to avoid the offset and achieve uniform, reliable welds.

3.4. Optimization through multiple overscans at low fluence

Based on weld seam quality results, optimal bonding conditions are identified as a combination of high repetition rate, low fluence per pulse, and a flexible range of scanning speeds. To achieve this, the maximum repetition rate of the existing system is utilized, and various scanning speeds are systematically investigated. While achieving a system without offset ideally requires very low fluence, bonding fails to occur at the low fluence levels. Slight increases in fluence introduce a vertical offset in the weld seam, complicating direct optimization. The proposed strategy involves using the maximum fluence that has no offset, regardless of whether bonding occurs, combined with a higher number of overscans. This approach leverages the fact that each laser pass alters the polymer's optical properties, particularly its transparency. Fig. 15 illustrates the progressive reduction in optical transparency of polycarbonate at the weld interface after multiple laser passes using the maximum offset-free fluence (0.08 J/cm^2) at a scanning speed of 10 mm/s. It reveals a consistent decrease in transmittance across the visible spectrum with increasing laser passes: from approximately 90% in pristine PC to around 20% after nine repetitions. This confirms that each pass incrementally reduces transparency, enhancing absorption and enabling bonding at lower fluences over time. Based on these observations, the described low-fluence, multi-pass approach establishes a robust processing regime that prevents pre-focus absorption, beam distortion, and thermal damage, while also significantly reducing oxygen-induced burning due to the low overall fluence. However, bonding does not occur at low fluences with a limited number of overscans; instead, multiple repetitions are required to build sufficient absorption. At 10 mm/s, bonding is achieved with a fluence of 0.08 J/cm^2 only after seven or more repetitions. At higher scanning speeds, optimal bonding parameters shift accordingly: five repetitions at 0.15 J/cm^2 for 20 mm/s, and seven repetitions at 0.17 J/cm^2 for 30 mm/s. The influence of the number of overscans can be understood through the evolution of energy absorption rather than fluence delivery. At low fluence, a single pass leaves the polymer largely transparent, but each repetition increases absorption and, in turn, the amount of energy deposited at the interface. As a result, weld strength rises progressively until a stable joint is achieved, beyond which additional passes provide little gain and may even trigger local overheating or structural irregularities. Thus, overscans enable controlled energy build-up for bonding, but only within an optimal window where strength is maximized without thermal damage.

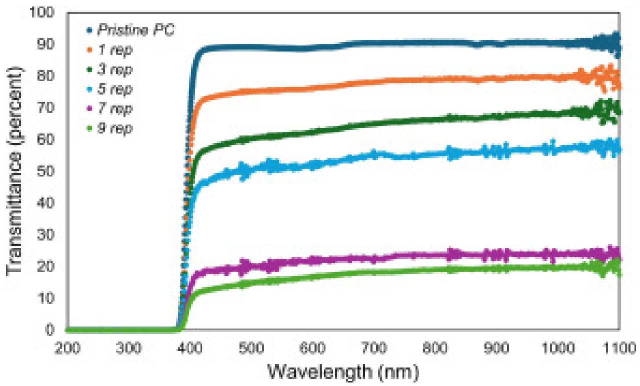

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Fig. 15. Transmittance spectra of polycarbonate at the weld interface after multiple laser passes using a fluence of 0.08 $\mathrm{J/cm^2}$ at a scanning speed of 10 $\mathrm{mm/s}$.

3.5. Chemical and mechanical characterization of optimized welds

Fig. 16 shows the Fourier-transform infrared (FTIR) spectra of pristine polycarbonate and processed PC samples obtained at a scanning speed of 10 mm/s and a fluence of 0.08 $J \cdot cm^{-2}$ with 1–9 overscans. All spectra are normalized to the aromatic $C = C$ stretching band at 1500–1515 cm^{-1}, which remains stable under processing and serves as an internal reference. For direct comparison, the integrated areas of the dominant peaks are calculated across conditions. The broad OH region (3200–3600 cm^{-1}) shows an increase with overscans, with the integrated area rising by over 45% from pristine to nine passes. The $H - O - H$ region (1620–1660 cm^{-1}) also increases by 34% over the same range, consistent with the accumulation of water-associated species on the processed surface. In contrast, the carbonate $C = O$ band (1745–1795 cm^{-1}) shows a decrease of 30%, whereas the $C - O$ related regions (1180–1260 cm^{-1} and 1000–1120 cm^{-1}) exhibit reductions of ~39% and ~23%, respectively. The reduction in the C—O absorption bands at 1180–1260 cm^{-1} and 1000–1120 cm^{-1} is consistent with a partial cleavage of carbonate and ether linkages along the polycarbonate backbone in the laser-modified region. Such reactions are expected to shorten the effective chain length, thereby locally reducing the average molecular weight. Molecular modelling and experimental studies on bisphenol-A polycarbonate and other amorphous polymers show that a decrease in molecular weight leads to a reduction of the glass transition temperature and a corresponding increase in segmental chain mobility [48], [49]. On this basis, the laser-affected region can reasonably be expected to exhibit higher chain mobility than the pristine bulk, which facilitates molecular interdiffusion and interpenetration across the interface between the two sheets and thereby supports the formation of a stronger welded junction. In contrast, regions away from the weld remain largely unaffected.

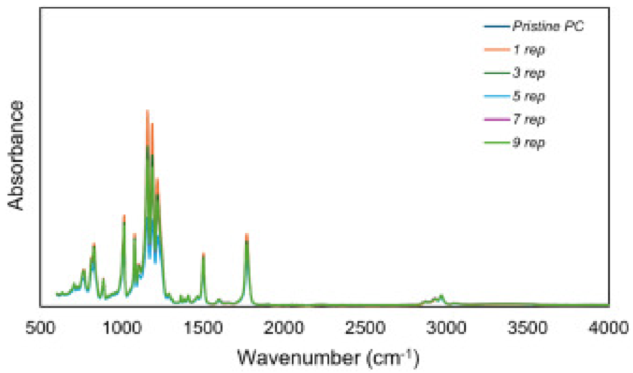

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Fig. 16. FTIR spectra of pristine polycarbonate and laser-processed samples (10 $\mathrm{mm \cdot s^{-1}}$, 0.08 $\mathrm{J \cdot cm^{-2}}$, 1–9 overscans).

This surface chemistry evolution directly correlates with the UV–Vis transmittance measurements (Fig. 15), which shows a monotonic reduction in transmission from ~90% in pristine PC to ~20% after nine passes. The formation of oxygen-containing species and the loss of C—O structural integrity introduce additional absorption pathways, progressively reducing transparency even though the nominal fluence per pass remains constant. Accordingly, subsequent laser passes experience higher effective absorption near the surface, explaining the observed shift in energy deposition toward the entrance region during multi-pass scanning.

Fig. 17 presents a series of optical microscope images showing laser weld seams formed under varying processing conditions using a femtosecond laser. Each row corresponds to an increasing number of overscans at a constant fluence and scanning speed, allowing direct observation of the effect of repetition on weld morphology. The columns represent increasing fluence levels (0.08 J/cm^2, 0.15 J/cm^2, and 0.17 J/cm^2) paired with scanning speeds of 10 mm/s, 20 mm/s, and 30 mm/s, respectively. As the number of overscans increases, the weld seams become progressively wider due to cumulative energy deposition. For each scanning speed, all tested overscan levels—9, 12, and 15 for 10 mm/s; and 5, 7, and 9 for both 20 mm/s and 30 mm/s—resulted in continuous and more uniform bonding. This demonstrates a broad and stable process window across all tested parameters. The observed variation in weld seam thickness provides flexibility in tailoring bond width to meet specific application requirements.

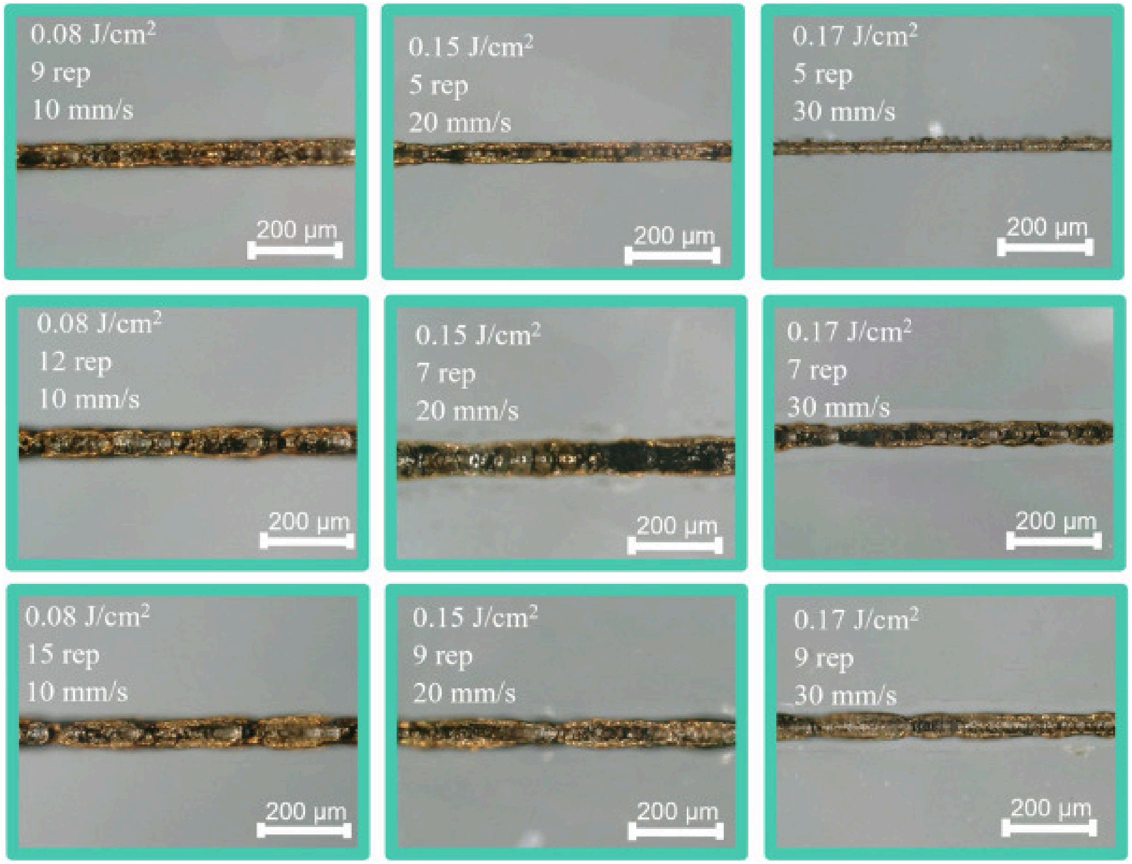

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Fig. 17. Weld seam regions for various pulse energies and numbers of overscans.

A representative cross-sectional optical micrograph of adjacent weld lines processed at 10 mm/s and 0.08 J/cm^2 with nine passes is presented in Fig. 18. This parameter set lies within the stable bonding window identified in previous sections and demonstrates a continuous fused region across the interface. No separation is visible between the two polycarbonate layers, and the processed zone extends across the contact plane, indicating localized interfusion. The initial laser focal position was set on the top surface of the bottom plate, which explains the larger processed volume on the lower side of the interface. Importantly, the cross-section is obtained while the two plates remain physically bonded, further supporting that the interface is continuously fused rather than simply in contact. This cross-section visually confirms interfacial fusion in the low-fluence multi-pass regime.

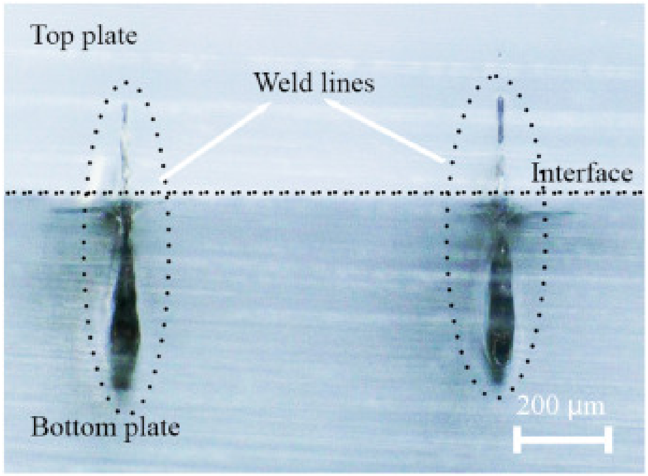

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Fig. 18. Cross-sectional optical image of welds processed at 10 mm/s and 0.08 J/cm^2 with nine passes, showing a continuous fused region across the interface.

Fig. 19 represents the shear strength of the samples created using high repetition and low fluence per pulse, with three samples tested for each setting. Three different laser parameter sets are plotted: 10 mm/s at 0.08 J/cm^2, 20 mm/s at 0.15 J/cm^2, and 30 mm/s at 0.17 J/cm^2. The data suggest that increasing the number of overscans initially increases shear strength, but it stabilizes or decreases beyond a certain point.

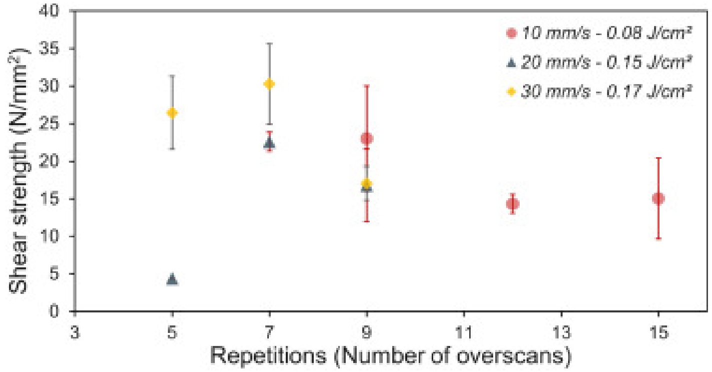

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Fig. 19. Shear strength as a function of the number of overscans for transparent polycarbonate samples processed with different parameter sets.

Except for the sample processed at 20 mm/s with 5 repetitions, all others exhibit a shear strength exceeding 15 N/mm^2. The samples processed at 30 mm/s with 7 and 5 repetitions show the highest strength, reaching 30.2 N/mm^2 and 26.4 N/mm^2, respectively. The yield strength of polycarbonate is approximately 62.5 N/mm^2 under quasistatic conditions [50]. The strongest bonded samples reach nearly 48% of this value, clearly demonstrating that the optimized parameters enable high-strength joints relative to the pristine material.

The Weibull distribution is a powerful statistical tool employed to quantify joint strength reliability, where the Weibull modulus (β) indicates process consistency and higher values represent lower variability. Fig. 20 shows the Weibull distribution analysis of three parameter sets: 10 mm/s with a fluence of 0.08 J/cm^2 and 12 overscans, 20 mm/s with a fluence of 0.15 J/cm^2 and 7 overscans, and 30 mm/s with a fluence of 0.17 J/cm^2 and 7 overscans. All datasets demonstrate strong correlation coefficients with R^2 values exceeding 0.92, confirming reliable Weibull fitting.

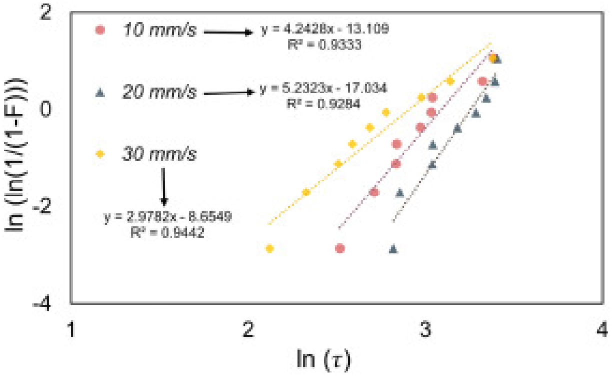

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Fig. 20. Weibull probability plots of shear stress data at scanning speeds.

For the 10 $mm\,/s$ condition, the Weibull modulus is 4.24 and the scale parameter is 21.25 $N/\,mm^2$, meaning that 63.2% of specimens are expected to fail at shear strengths below this value. The 20 $mm\,/s$ condition exhibits the highest Weibull modulus of 5.23 and a superior scale parameter, with the characteristic strength at 25.7 $N/\,mm^2$ and excellent process consistency. At 30 $mm\,/s$, both strength and process consistency are reduced, with the Weibull modulus of 2.96 and scale parameter of 18.17 $N/\,mm^2$, corresponding to a broader strength distribution and higher process variability. All three datasets show correlation coefficients R^2 > 0.92, confirming reliable Weibull fitting. These results demonstrate that the 20 $mm\,/s$ condition provides an optimal balance of laser processing parameters, yielding the most reliable welding process for transparent polymer joining with superior consistency and the highest characteristic strength. The Weibull statistical analysis reveals systematic differences in both joint strength and process variability as functions of welding parameters, confirming the critical importance of parameter optimization for reliable ultrashort laser welding.

To demonstrate the practical application of the proposed femtosecond laser welding technique, a microfluidic device is fabricated by bonding an injection-molded polycarbonate substrate containing microchannels to a flat polycarbonate cover plate. Based on the optimization results, scanning speeds of 30 and 10 mm/s provided the best outcomes in terms of shear strength, while Weibull distribution analysis highlighted 20 and 10 mm/s as the most reliable conditions. Taking both strength and reliability into account, a scanning speed of 10 mm/s is selected as the processing parameter for microfluidic channel fabrication. Welding is performed at a scanning speed of 10 mm/s with an energy density of 0.08 J/cm^2, employing 12 overscans and a lateral offset of 0.2 mm to ensure consistent seam formation. The sealed microfluidic chip undergoes leakage tests. A static pressure decay test is conducted by pressurizing the device to 1.0 bar and holding this pressure for 5 min; no pressure drop is observed, confirming leak tightness under dynamic flow conditions. Visual inspection and dye-filling assays confirm complete filling of the microchannels without leakage, as shown in Fig. 21. Three devices exhibit consistent performance, demonstrating the robustness and reproducibility of the femtosecond laser welding process.

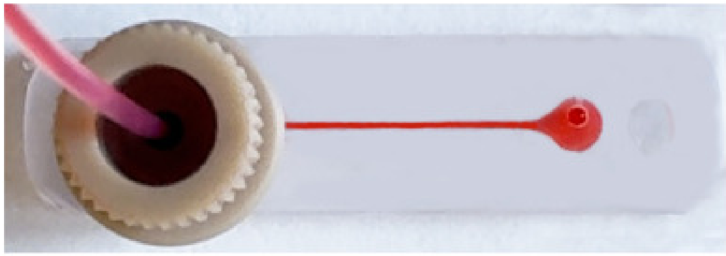

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Fig. 21. Leakage test setup of a welded polycarbonate microfluidic chip with dyed liquid flowing through the sealed microchannel under 1 bar pressure.

4. Conclusion

This study investigated the bonding of polycarbonate using femtosecond laser pulses, beginning with an analysis of how fluence per pulse, scanning speed, and repetition rate influenced weld seam morphology. It was found that material degradation occurred due to two main factors: high fluence per pulse at high scanning speeds, and high cumulative fluence at lower scanning speeds. During this investigation, a spatial offset between the intended focal plane and the actual processing zone was identified. Although the focus was pre-adjusted to align the beam with the polymer interface, irregularities in weld seam morphology became more pronounced compared to weld seams obtained under identical parameters without focus pre-adjustment, where the processing area remained within the top plate. These issues were attributed to beam profile distortion at the offset position. To understand this effect, optical simulations were performed in Zemax OpticStudio, which revealed that the beam shape is significantly altered outside the ideal focal plane, leading to inconsistent energy deposition and degraded weld quality. These insights deepened the understanding of ultrafast laser-material interactions outside the focal plane, where even slight variations can strongly influence absorption, energy delivery, and process stability due to the high peak power involved. To address these issues, a welding strategy was proposed that uses low pulse energies, normally insufficient for bonding, combined with a high number of overscans. This method produced cleaner weld seams and improved reliability by minimizing hard-to-control variables linked to the nonlinear nature of the process. FTIR measurements confirmed that no major chemical degradation occurs in the bonded regions. The highest shear strength of 30.2 N/mm^2, corresponding to 48% of the strength of pristine polycarbonate, was achieved at a scanning speed of 30 mm/s with 7 overscans, demonstrating the effectiveness of the strategy. Weibull analysis was performed for three different scanning speeds, with the most promising result obtained at 20 mm/s, 0.15 J/cm^2, and 7 repetitions, yielding a Weibull modulus of 5.2. Based on an evaluation of both strength and reliability, a scanning speed of 10 mm/s was adopted for the fabrication of microfluidic channels. Bonded chips produced under these conditions successfully withstood 1 bar internal pressure during leakage testing, confirming sealing reliability under practical operating conditions. This method effectively eliminated offset issues induced by high cumulative fluence and fluence per pulse. By avoiding the high fluence regime, which was a primary limiting factor for achieving higher scanning speeds, this strategy opened the possibility for faster processing, which will be explored in future investigations.

Declaration of competing interest

The authors declare no competing interest.

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